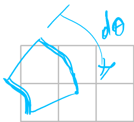
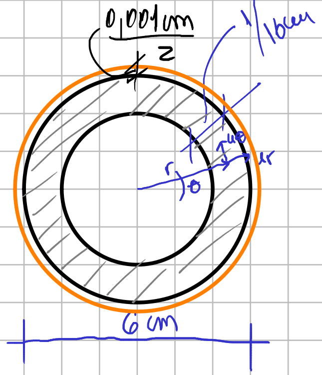
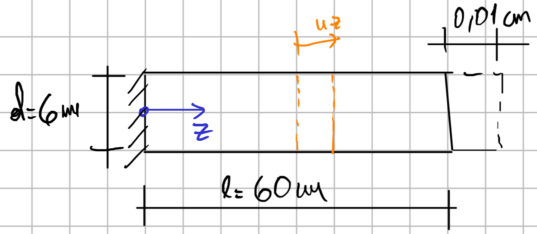
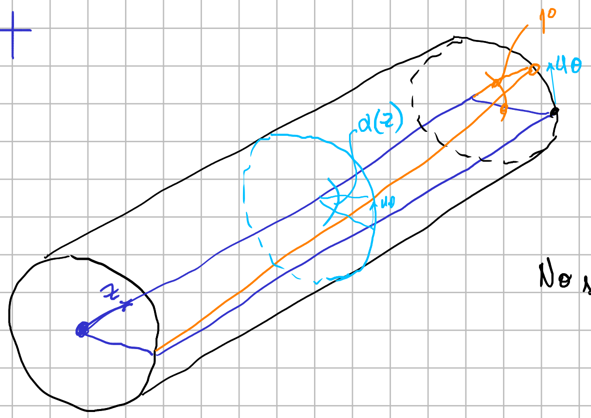


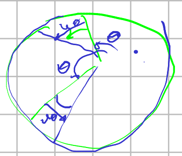


Deformación uniforme



Pequeños deformaciones





No se produce alabeo

1) Estirado axialmente 0,010 cm
2) Expansión en el diámetro 0,001 cm
3) Giro de torsión 1°

→ Desde los desplazamientos se quiere obtener las deformaciones.

$$e_{rr} = \frac{\partial u_r}{\partial r}, \quad \times$$

$$e_{\theta\theta} = \frac{u_r}{r} + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta}, \quad \times$$

$$e_{r\theta} = \frac{1}{2}\left(\frac{1}{r}\frac{\partial u_r}{\partial \theta} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r}\right), \quad \times$$

$$e_{zr} = \frac{1}{2}\left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r}\right),$$

$$e_{z\theta} = \frac{1}{2}\left(\frac{1}{r}\frac{\partial u_z}{\partial \theta} + \frac{\partial u_\theta}{\partial z}\right), \quad \times$$

$$e_{zz} = \frac{\partial u_z}{\partial z}. \quad \times$$

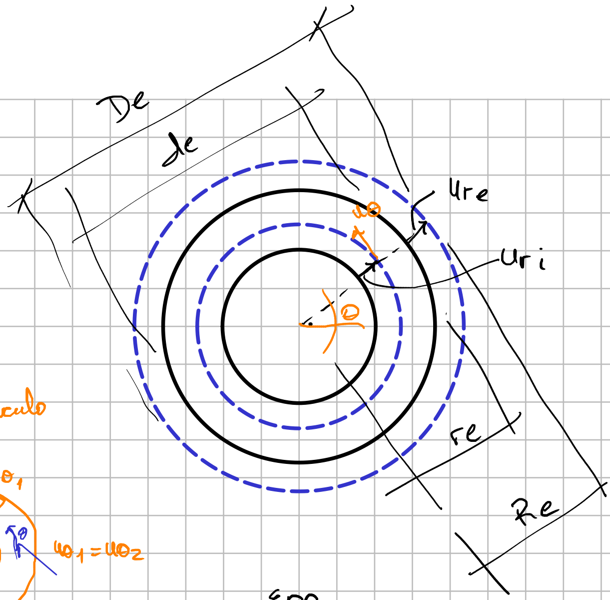
- ε_{zz}

deformación uniforme

$$\varepsilon_{zz} = \frac{\partial u_z}{\partial z} \Rightarrow u_{z_{max}} - u_z(0) = \int_0^l \varepsilon_{zz}\, dz = \varepsilon_{zz}\, z\Big|_0^l = \varepsilon_{zz}\, l$$

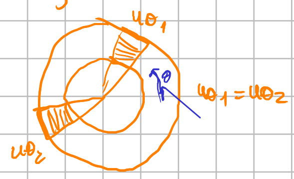
$$\varepsilon_{zz} = \frac{\overbrace{u_z - u_z(0)}^{\Delta l}}{l} = \frac{0,01\,cm - 0\,cm}{60\,cm} = 1,67 \times 10^{-4}$$

• $\varepsilon_{\theta\theta}$

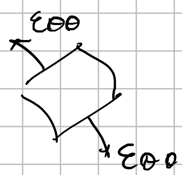


$$\varepsilon_{\theta\theta} = \frac{u_r}{r} + \frac{1}{r}\cdot\frac{\partial u_\theta}{\partial \theta}$$

$$\frac{\partial u_\theta}{\partial \theta} = 0$$ → no cambia con θ al girar como un círculo alrededor de su eje



$$\varepsilon_{\theta\theta} = \frac{u_r}{r}$$

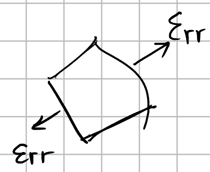


$$\varepsilon_{\theta\theta e} = \frac{u_{re}}{r_e} = \frac{\frac{0{,}001\,cm}{2}}{\frac{6\,cm}{2}} = 1{,}67\times10^{-4}$$

$$\frac{\cancel{\pi} De - \cancel{\pi} de}{\cancel{\pi} de} = \frac{De - de}{de} = \frac{\cancel{2}R_e - \cancel{2}r_e}{2r_e} = \frac{\overbrace{R_e - r_e}^{u_{re}}}{r_e} = \frac{u_{re}}{r_e} = \varepsilon_{\theta\theta}$$

$$\varepsilon_{\theta\theta i} = \frac{u_{ri}}{r_i} = \frac{\frac{0{,}001\,cm}{2}}{\left(6\,cm - \frac{1}{16}\,cm\right)} = 1{,}70\times10^{-4}$$

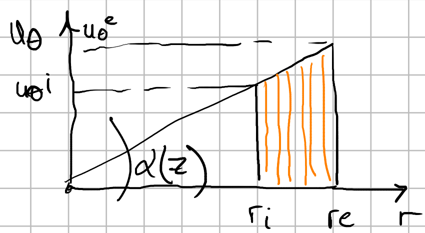
• ε_{rr}



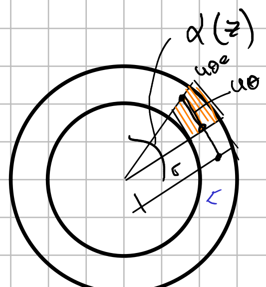
$$\varepsilon_{rr} = \overset{\frac{\Delta u_r}{\Delta r}}{\frac{\partial u_r}{\partial r}} = \frac{u_{re} - u_{ri}}{\underbrace{r_e - r_i}_{\text{longitud inicial}}} = \frac{0}{1/16\,cm} = 0$$

→ y $u_{re} = u_{ri}$

• $\varepsilon_{r\theta}$



$$\varepsilon_{r\theta} = \frac{1}{2}\left(\frac{1}{r}\overset{0}{\frac{\partial u_r}{\partial \theta}} + \frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r}\right)$$



$$\alpha(z) \simeq tg\,\alpha(z) = \frac{u_\theta}{r} \Rightarrow u_\theta = \alpha(z).r$$

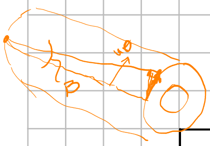
$$\frac{\partial u_\theta}{\partial r} = \alpha(z).\frac{\partial r}{\partial r} = \alpha(z)$$

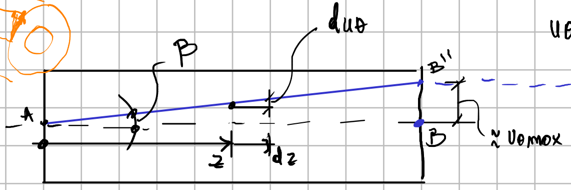
$$\frac{u_\theta}{r} = \frac{\alpha(z).\cancel{r}}{\cancel{r}} = \alpha(z)$$

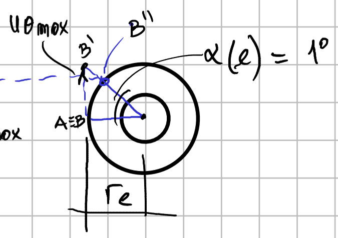
$$\varepsilon_{r\theta} = \frac{1}{2}\left(\frac{\partial u_\theta}{\partial r} - \frac{u_\theta}{r}\right) = \frac{1}{2}\left(\alpha(z) - \alpha(z)\right) = 0$$

• $\varepsilon_{z\theta}$

$$\varepsilon_{z\theta} = \frac{1}{2}\left(\frac{1}{r}\cdot\overbrace{\frac{\partial u_z}{\partial \theta}}^{0} + \frac{\partial u_\theta}{\partial z}\right) = \frac{1}{2}\cdot\frac{\partial u_\theta}{\partial z}$$







$$tg\,\beta = \frac{du_\theta}{dz} = \frac{u_\theta\, max}{\ell}$$

$$tg(\alpha(\ell)) = \frac{u_\theta max}{re} \Rightarrow$$

$$\Rightarrow u_\theta max = tg(\alpha(\ell)).re$$

$$\frac{du_\theta}{dz} = \frac{tg(\alpha(\ell)).re}{\ell}$$

$$\varepsilon_{z\theta} = \frac{1}{2}.\frac{du_\theta}{dz} = \frac{1}{2}\frac{tg(\alpha(\ell)).re}{\ell} = \frac{1}{2}.\frac{tg(1°).\frac{6cm}{2}}{60cm} = 4{,}36\times10^{-4}$$

• ε_{zr}

$$\varepsilon_{zr} = \frac{1}{2}\left(\overbrace{\frac{\partial u_r}{\partial z}}^{0} + \overbrace{\frac{\partial u_z}{\partial r}}^{0}\right) = 0$$